



Learning from interactive video: the influence of self-explanations, navigation, and cognitive load

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Abstract

Recent research has shown that enhancing instructional videos with questions, such as self-explanation prompts, and thus shifting the process from receptive to constructive learning, is beneficial to learning. However, the inclusion of questions is often confounded with the implementation of learner pacing through navigation features. Furthermore, previous studies have often not controlled for learning time. To address these shortcomings, an experiment ($N=128$) was conducted. Participants watched an instructional video about cloud formation and lightning, with learning time controlled. In a 2×2 between-subjects design, navigation features (learner pacing vs. system pacing) and self-explanation prompts (prompts vs. no prompts) were manipulated. The results showed no effects of navigation features and self-explanation prompts on learning performance. While navigation features did not affect cognitive load, self-explanation prompts increased both intrinsic and extraneous cognitive load. Overall, the quality of responses to prompts was low but positively related to comprehension. The results are discussed in terms of the Interactive-Constructive-Active-Passive framework and Cognitive Load Theory. They highlight the importance of boundary conditions when investigating the effects of interactive features in instructional videos.

Keywords Interactive video · Self-explanation · Navigation · Cognitive load · Educational effectiveness

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Introduction

The use of videos in education is a growing trend, as observed by Fyfield et al. (2019) and Moreno-Guerrero et al. (2020). With the increasing popularity of instructional videos, several researchers have reviewed their impact on learning such as Brame (2016), Noetel et al. (2021), Poquet et al. (2018), and Sauli et al. (2018). For example, Noetel et al. (2021) highlighted that replacing traditional teaching methods with instructional videos resulted in only marginal improvements in learning performance (effect size $g=0.28$). Brame (2016) argued that videos have no inherent educational effectiveness, emphasizing the importance of instructors using design principles to make learning from videos effective.

In line with the Interactive-Constructive-Active-Passive (ICAP) framework proposed by Chi and Wylie (2014), it is suggested that learners should actively engage with videos rather than simply watching them. This is consistent with Cognitive Load Theory (CLT; Paas & Sweller, 2022), which suggests that efforts should be made during learning to enhance germane processes such as the acquisition of cognitive schemas. One way to promote learner engagement with videos is to extend them with interactive features. Sauli et al. (2018) distinguished between navigation features and enhanced features in instructional videos. Navigation features can include controls such as play and pause buttons, a table of contents, and an index. They allow learners to navigate within the video and to access additional resources. Enhanced features can include prompts and questions to support retention and comprehension of information presented in the video. In a recent meta-analysis, Ploetzner (2024) found that instructional videos with enhanced interaction features significantly improved learning (effect size $g=0.52$).

However, in many of the studies analyzed by Ploetzner (2024), the videos were augmented with both enhanced and navigation features. This raises the question of which of these features contributed to successful learning. The inclusion of prompts and questions may not only enhance germane processes, but also increase extraneous cognitive load, which may impede learning. Specifically, responding to prompts and answering questions disrupts the flow of information and thus might hinder the development of cognitive schemas. Navigation features may enhance germane processes by allowing learners to actively navigate in the video according to their information needs. At the same time, however, this requires learners to engage in self-regulatory and even motor processes (Sepp et al., 2019), which may increase cognitive load and thus hinder learning (Scheiter, 2022). In addition, about half of the studies analyzed by Ploetzner (2024) either did not control for learning time or left its control unclear. According to a recent review by Godwin et al. (2021), although the relationship between learning time and learning performance is still not well understood, many studies report positive correlations between the two variables. Thus, learning time is a potentially confounding factor in previous research on learning from interactive video. This raises the question of whether the learning effects observed in previous research are due to the availability of interaction features or to differences in learning time.

To disentangle the aspects described above, we report an experimental study of learning from instructional video in which two factors were varied: presentation of self-explanation prompts and availability of navigation features. Learning time was kept constant in all conditions. Learners' prior knowledge and learning performance were assessed. Learners were also asked to rate the cognitive load induced by the learning task. The next section summarizes the main theories and empirical findings underlying the study. The research questions

and hypotheses are then presented. Next, the study and its results are described. A discussion and conclusions complete the paper.

Theory and empirical findings

The ICAP framework

The ICAP framework is a theoretical model proposed by Chi and Wylie (2014, see also Chi et al., 2018) that categorizes students' cognitive engagement based on their observable behaviors during learning. The framework predicts that deeper engagement and more effective learning outcomes are achieved when students move from passive to more interactive behaviors. In passive mode, learners receive information without doing anything related to learning beyond attending to the material. Examples include listening to a lecture without taking notes or watching a video without interacting. The ICAP framework assumes that learning in this mode occurs through the isolated storage of information, often resulting in inert knowledge that is difficult to apply in new contexts. In active mode, learners engage in some form of physical manipulation or motor activity that focuses attention on the learning material, such as taking verbatim notes or navigating within a video. This mode activates prior knowledge and allows learners to integrate new information with what they already know, filling in gaps and reinforcing existing knowledge schemas. In constructive mode, learners generate additional output beyond what is provided in the learning materials, such as self-explanations or summaries in their own words. Constructive behaviors involve inferring new knowledge from existing information, resulting in enriched and more coherent knowledge structures that support deeper understanding and transfer of learning. In interactive mode, learners engage in dialogic interactions where both parties contribute constructively, such as discussing and debating ideas or asking and answering questions. This mode requires mutual and frequent exchange of ideas, where learners build on each other's contributions, resulting in the highest levels of learning, including the potential for innovative thinking and deep understanding.

The framework has found support in empirical studies (e.g., Lim et al., 2019; Pi et al., 2023), has been adapted to different educational settings (e.g., Quesnelle et al., 2021), but has also faced critical discussion due to inconsistent findings, particularly regarding the educational effectiveness of the proposed sequence of engagement modes (e.g., El-Mansy et al., 2022; Qi et al., 2022).

Since the focus of the current study is on individual learning, three modes of the ICAP framework are implemented: the passive mode by watching the video, the active mode by providing the learners with navigation features, and the constructive mode by prompting the learners to construct self-explanations based on questions. In a further condition, learners are provided with both navigation features and self-explanation prompts. While the ICAP framework predicts that navigation features and self-explanation prompts enhance learning, it's important to acknowledge that negative effects are also possible. This is where CLT comes into play.

Cognitive Load Theory

The CLT (Paas & Sweller, 2022; Sweller et al., 1998) distinguishes three sources of cognitive load: intrinsic, germane, and extraneous cognitive load. Intrinsic cognitive load (ICL) is determined by the learner's prior knowledge and the complexity of the information being presented, referred to as element interactivity. When element interactivity is high, it requires the simultaneous processing of many elements in working memory, and thus can overload it. As emphasized by Kirschner (2002), ICL plays a central role in the construction of cognitive schemas in long-term memory. Germane cognitive load (GCL) is characterized by the mental effort invested in managing ICL and constructing cognitive schemas (cf. Paas & van Merriënboer, 2020). Finally, extraneous cognitive load (ECL) arises from the way in which the learning material is designed and presented to the learner (cf. Sweller et al., 2019). If information is presented in an unfavorable manner, it may require the learner to engage in cognitive processes that are not relevant to learning. Therefore, it is important to minimize ECL.

On the one hand, according to the ICAP framework, navigation features and enhanced features such as self-explanation prompts in instructional videos can increase GCL and thus improve learning. Providing learners with navigation features is often considered an effective way to counteract the transient information effect during learning from dynamic visualizations (Jiang & Sweller, 2022). This effect describes the observation that the continuous flow of transient information can overwhelm the learners and hinder learning. Navigation features, in contrast, allow learners to tailor the presentation to their information needs and give the learners more time to process the presented information (cf. Chen & Yen, 2021; Moritz et al., 2020; Rey et al., 2019). This should be reflected in enhanced GCL scores. On the other hand, navigation in a video requires learners to engage in self-regulatory and even motor processes (Sepp et al., 2019). This can be considered a form of ECL that may increase cognitive load and hinder learning (Scheiter, 2022). This is further supported by Renkl and Scheiter (2017), who found that effective use of navigational features requires metacognitive knowledge of when to pause an instructional video. Thus, students might stop the video at sub-optimal positions (e.g., in the middle of an important explanation) which may lead to an incomprehensive mental representation of the explanatory structure. Lowe (2008) supported this argument by investigating log-file data. He found that learners did not make effective use of navigation features and mainly used them for exploring rather unimportant and superficial aspects of an animation.

Similarly, self-explanation prompts can induce deeper information processing (Dunlosky et al., 2013), which increases GCL and improves learning (Ploetzner, 2024). However, this assumption is not fully supported by empirical investigations. Several studies report no significant benefits of prompts and questions on learning processes. For example, Zheng et al. (2022) found that prompting learners to self-explain did not result in significantly more engagement with the video than learning without prompting. Hefter et al. (2023) also found that prompting did not significantly affect learning from video, even though it elicited self-explanations. Moreover, when learners are asked to answer questions, the video typically pauses. This interrupts the flow of information. After providing answers, learners may need to re-establish their information processing, which may interfere with schema construction. Consequently, self-explanation prompts might even induce additional ECL and thus, learning might be hindered.

Research questions and hypotheses

In contrast to previous research, the current study aimed to examine both the separate and combined effects of navigation features (e.g., play/pause button) and enhanced interaction features (self-explanation prompts) on learning from instructional video while holding learning time constant. The following research questions were addressed:

1. Is learning from video with navigation features more successful than learning from video without navigation features? Since a large body of research supports navigation features in dynamic visualizations (e.g., Rey et al., 2019), it was hypothesized that these features would be beneficial for learning.
2. Is learning from video with self-explanation prompts more successful than learning from video without self-explanation prompts? In accordance with the meta-analysis by Ploetzner (2024) and the review by Fonseca and Chi (2011), it was hypothesized that self-explanation prompts would improve learning from video.
3. Do the availability of navigation features and self-explanation prompts interact with the success of learning from video? Self-explanation prompts are intended to induce constructive processes such as providing justifications and drawing inferences (Chi, 2022). Learners' engagement in such processes may lead to the identification of knowledge gaps and the desire to fill these gaps by revisiting previously viewed parts of the video using navigation features. It was therefore hypothesized that navigation features are particularly helpful in combination with self-explanation prompts.
4. Do more complete and accurate self-explanations lead to more successful learning? Consistent with the findings of Hefter and Berthold (2023) and Hefter et al. (2023), it was hypothesized that the quality of responses to self-explanation prompts is positively related to learning success.
5. Do navigation features and self-explanation prompts affect cognitive load? It was hypothesized that self-explanation prompts would increase germane cognitive load due to their very design. In addition, as described above, it was hypothesized that navigation features and self-explanation prompts would increase extraneous cognitive load.
6. Are the effects of navigation features and self-explanation prompts mediated by cognitive load? To better understand the relationship between the investigated design features and learning processes, an exploratory approach to this research question is taken using mediation analysis. No specific hypothesis was outlined because of the exploratory approach.

Method

Design

In a 2×2 between-subjects design, two factors were varied: presentation of self-explanation prompts (no vs. yes) and availability of navigation features (no vs. yes). Learners' prior knowledge and learning performance were assessed. Furthermore, immediately after learn-

ing, the learners had to rate the intrinsic, extraneous, and germane cognitive load induced by the learning task.

Participants

The meta-analysis by Ploetzner (2024) reported a medium effect size of $g=0.52$ for the effect of enhanced interactive features in instructional videos. With respect to this effect size, an a priori power analysis for a 2×2 analysis of variance with $\alpha=0.05$ and a power of $1-\beta=0.80$ indicated that 118 participants should be studied. A total of 128 students (80.5% female, mean age $M=22.03$ years, $SD=3.40$) at a university in Germany volunteered for the study. They received financial compensation for their participation. Students were randomly assigned to one of the four experimental conditions.

Instructional video and self-explanation prompts

Based on various online materials for middle school physics education, an instructional video about how thunderstorms and lightning work was created by using the Tumult Hype (tumult.com) animation software. The video is divided into three sections: cloud formation, ice crystal formation and rain, and charge separation and lightning. It presented animations of the various phenomena combined with a voice-over narration. It lasted four minutes and nine seconds.

In a meta-analysis, Bisra et al. (2018) found that self-explanation prompts are particularly effective when the learners have little prior knowledge, the prompts are specific rather than nonspecific, and the self-explanations require learners to draw inferences (see also Berthold et al., 2009; Fiorella & Mayer, 2015). Accordingly, specific self-explanation prompts were designed that consisted of two parts. First, an open question. For example, “Why do large, negatively charged crystals collect in the lower part of the cloud?” Second, to guide learners in formulating an answer, a sentence opener that had to be completed with an explanatory statement. For example, “Large, negatively charged crystals collect in the lower part of the cloud because ...”. Answers required learners to combine multiple pieces of information previously given in the learning material and to draw inferences. Overall, six self-explanation prompts with an open-answer format were included.

In accordance with the experimental design described above, four variations of the video were created using the video platform PlayPosit (go.playposit.com):

1. The video was presented without self-explanation prompts and without navigation features. In this variant, learners could watch the video and restart it at the end to watch it again.
2. The video was presented with six self-explanation prompts, but without navigation features. In this variant, learners could watch the video and had to answer self-explanation prompts. Within the overall learning time, learners could take as much time as they needed to answer the questions. At the end of the video, learners could restart it. Previously answered questions could be reviewed and revised while the video was being replayed. In addition, before the video was presented, learners received instruction on what self-explanations are and how they can benefit learning.

3. The video was presented without self-explanation prompts, but with standard navigation features. These allowed learners to pause and restart the video, jump 10 s forward or backward in the video, and jump freely to any location within the video using a scrub bar slider. In this variant, learners could watch the video, navigate freely within the video, and start again at the end. In accordance with Lin et al. (2022), before the video was presented, learners received instruction on how to improve their learning by strategically using navigation features.
4. The video was presented with self-explanation prompts and navigation features. This variant combines the second and third variants, including the instructions described above. In particular, learners were able to navigate within the video while answering questions and review previously answered questions at any time. Again, learners were able to review and revise the self-explanation prompts.

Figure 1 shows a screenshot of the video with self-explanation prompts and navigation features. The dots on the scrub bar slider indicate locations where self-explanation prompts have occurred. Self-explanation prompts automatically opened when a corresponding location was reached. Figure 2 displays a screenshot of the same video at a slightly later point in time with an opened self-explanation prompt.



Fig. 1 Screenshot of the video with self-explanation prompts and navigation features. Currently, all self-explanation prompts are closed, but their locations are indicated by the dots on the scrub bar slider

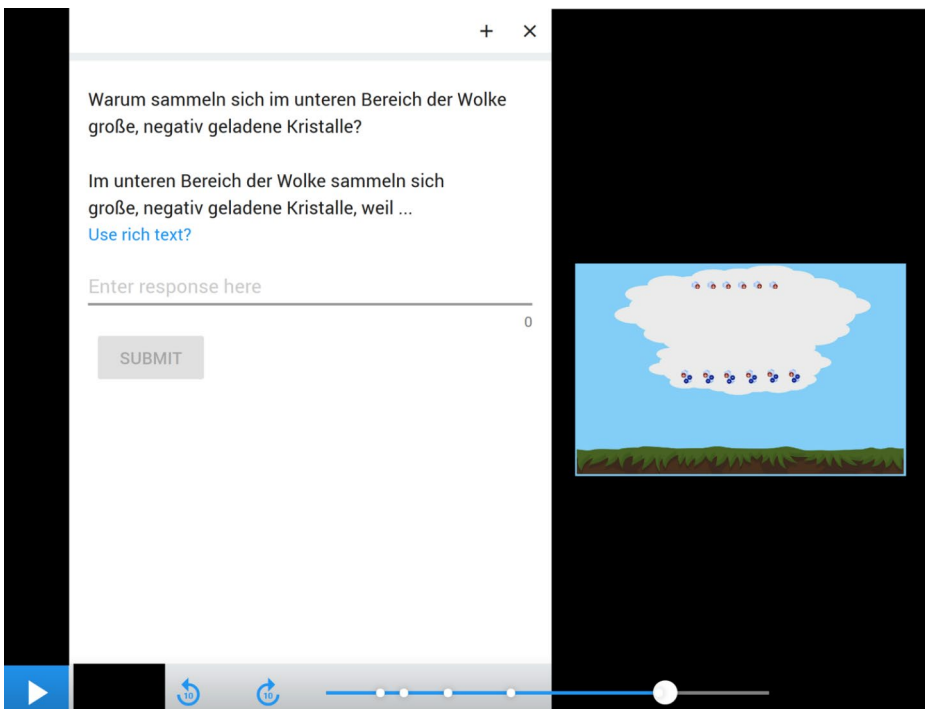


Fig. 2 Screenshot of the same video with an opened self-explanation prompt: Why do large, negatively charged crystals collect in the lower part of the cloud? Large, negatively charged crystals collect in the lower part of the cloud because ... (translation by the authors)

Assessments

Prior knowledge

Learners' prior knowledge of how thunderstorms and lightning work was assessed by six single-choice questions with four possible answers each, one correct and three incorrect. To ensure that earlier questions did not contain information that could be used to answer later questions, the questions were presented to the students in a fixed order. The order of answers in each question was randomized. Each correct answer was worth one point. An Item Response Theory (IRT) analysis (one-factor model) was conducted to assess the difficulty and discrimination parameters of all items. Results revealed that one item had a negative discrimination parameter and was thus removed from the statistical analysis. Five items remained and a maximum of five points could be achieved. The IRT characteristics of the remaining items are provided in Appendix A.

Learning performance

Learning performance was assessed by 24 single-choice questions with four possible answers each, one correct and three incorrect. While 12 questions assessed retention, 12 questions assessed comprehension. Retention questions required learners to remember

information directly given in the learning material. Comprehension questions required learners to combine multiple pieces of information given in the learning material and to draw inferences. Six comprehension questions addressed the same concepts addressed in the six self-explanation prompts but were not identical to them. Furthermore, six comprehension questions addressed concepts that were present in the video but not part of the self-explanation prompts. Again, to ensure that earlier questions did not contain information that could be used to answer later questions, the questions were presented to the students in a fixed order. The order of answers in each question was randomized. Each correct answer was worth one point.

An IRT analysis was conducted for the retention test (one-factor model) and the comprehensions test (two-factor model, since two subscales were created). Regarding the retention test, two items were removed due to a negative discrimination parameters. The same applied for the comprehension test (one for the questions that were close and one for the questions that were not close to the self-explanations prompts). The difficulty parameters revealed satisfactory values, indicating that a broad spectrum of questions with rather low, medium and high difficulty were implemented. The IRT characteristics of the remaining items are provided in Appendix A. A maximum of 20 points could be achieved (10 for retention and 10 for comprehension).

Cognitive load

Although attempts have been made to objectively and reliably assess cognitive load using neurophysiological and dual-task methods (for reviews see Brünken et al., 2003; Korbach et al., 2017), these methods are difficult to apply in educational settings. Therefore, in most cases, cognitive load is assessed using subjective ratings, despite their limitations in terms of objectivity and reliability (for reviews see Krieglstein et al., 2022; Paas et al., 2003). In this study, cognitive load was assessed using multiple scales as proposed by Krieglstein et al. (2023). Specific items were selected from the original scales of Krieglstein et al. (2023) to avoid redundancies and to maintain the focus on the objective and medium of the study. Two items assessed intrinsic cognitive load ($\alpha=0.52$; “The learning material was difficult to understand.”, “The learning content was complex.”), four items extraneous cognitive load ($\alpha=0.77$; “The design of the learning material made it difficult to find relevant information quickly.”, “It was difficult to gain an overview of the structure of the learning material.”, “The design of the learning material made it difficult to recognize links between individual information units.”, “Because of the design of the learning material, I had the impression that I could not concentrate on the learning content.”), and three items germane cognitive load ($\alpha=0.70$; e.g., “I made an effort to understand the learning content.”, “I actively reflected upon the learning content.”, “I was able to expand my prior knowledge with the learning content.”). Learners were asked to rate each item on a Likert scale ranging from 1 (does not apply at all) to 9 (applies to a full extent).

A pre-study was conducted to test the comprehensibility of all the material and the appropriateness of the learning time. It was also checked whether the study could be completed without technical problems. Therefore, 12 university students completed the entire experiment. At the end of the experiment, students were encouraged to share their experiences and suggest improvements. As a result of the pre-study, minor changes were made to clarify the written instructions before watching the instructional videos. In addition, the learning

time was increased from 9 to 12 min because the pre-study showed that, especially in the conditions with self-explanation prompts, there was not enough time to watch the video and answer the prompts carefully.

Procedure

Students were studied in groups of up to eight. Each student learned individually using a computer equipped with a 21-inch screen, a mouse, and over-ear headphones. The computers were placed on separate tables. All material was presented to the students using a computer program created with the authoring software ActivePresenter (atomisystems.com). The presentation was shown in full screen mode with a size of 1200×1000 pixels.

Initially, the students were told how to use the program and what they would be doing during the study. They were then asked for demographic information such as age, gender, and field of study. The program then presented the material to the learners in the following order: (1) prior knowledge assessment, (2) instructional video, (3) cognitive load assessment, and (4) learning performance assessment. The time for learning from the video was limited to 12 min in all four experimental conditions. A time bar at the bottom of the screen indicated the amount of time remaining to the learner. After the learning assessment, learners were given feedback on their performance score. The procedure took about 45 min.

Results

All analyses were carried out using JASP (JASP Team, 2023). Multivariate (MANOVA) and univariate (ANOVA) analyses of variance were used to assess differences between groups. Assumptions of these methods were tested and are reported only if they were significantly violated. If variance homogeneity was violated, ROBUST ANOVAs were performed. When ANOVAs yielded nonsignificant results, additional Bayes factors were examined to make more detailed statements about the probability of the null hypotheses. In this sense, the BF_{10} was performed and interpreted according to the suggestions of Jeffreys (1961). Consequently, a BF_{10} below 0.33 was interpreted as substantial evidence in favor of the null hypothesis.

Finally, mediation analyses were conducted to gain deeper insight into the potential mediating effect of cognitive load on learning performance. Therefore, the implementation of self-explanation prompts and navigation features were used as predictors, all facets of cognitive load were used as mediators, and retention and comprehension scores were used as exogenous variables in a path analysis. Descriptive data for all dependent variables are presented in Table 1.

Learner characteristics

Participants did not differ between the experimental groups regarding age, semester, and prior knowledge, $F = [1.89, 1.19, 1.24]$, $p = [0.136, 0.319, 0.300]$ as well as gender and field of study $\chi^2 = [4.10, 26.85]$, $p = [0.176, 0.663]$.

Hypothesis 1 Effect of navigation features on learning performance.

Table 1 Descriptive data for all groups

	Self-explanation prompts				No self-explanation prompts			
	Navigation features		No navigation features		Navigation features		No navigation features	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Prior knowledge	1.90	(38%)	2.28	(46%)	1.97	(39%)	2.25	(45%)
Retention	6.65	(67%)	6.47	(65%)	6.16	(62%)	6.75	(68%)
Comprehension	5.45	(55%)	5.00	(50%)	5.03	(50%)	5.63	(56%)
Close to prompts	2.65	(53%)	2.44	(49%)	2.33	(47%)	2.59	(52%)
Not close to prompts	2.81	(56%)	2.56	(51%)	2.70	(54%)	3.03	(61%)
Intrinsic cognitive load	4.76	(53%)	4.63	(51%)	4.00	(44%)	3.84	(43%)
Extraneous cognitive load	3.29	(37%)	3.81	(42%)	2.58	(29%)	2.65	(29%)
Germane cognitive load	7.93	(90%)	7.95	(88%)	8.18	(91%)	7.88	(88%)
Rating of self-explanations	2.70	(45%)	2.38	(40%)	0.97			

Note. Prior knowledge score ranged from 0 to 5 with a guessing probability of 25%. Retention and comprehension scores ranged from 0 to 10 with a guessing probability of 25%. Comprehensions cores that represent questions that were close and not close to the self-explanations ranged from 0 to 5 with a guessing probability of 25%. Cognitive load ratings ranged from 1 to 9. Self-Explanation ratings ranged from 0 to 9

A MANOVA with retention and comprehension scores as depended variables revealed no significant main effect for navigation features (*Wilk's* $\Lambda=1.00$, $F(2, 123)=0.04$, $p=.962$). Bayes factors indicated support for the null-hypothesis regarding retention ($BF_{10}=0.19$) and comprehension ($BF_{10}=0.19$). A descriptive plot is displayed in Fig. 3 (retention) and Fig. 4 (comprehension).

An additional MANOVA using the two types of comprehensions scores (thematically close vs. not close to the self-explanation prompts) revealed no significant main effect for navigation features (*Wilk's* $\Lambda=1.00$, $F(2, 123)=0.03$, $p=.967$). Bayes factors indicated support for the null-hypothesis regarding questions that were thematically close to the self-explanation prompts ($BF_{10}=0.19$) and not thematically close ($BF_{10}=0.19$). Hypothesis 1 is therefore not supported by the results.

Hypothesis 2 Effect of self-explanation prompts on learning performance.

A MANOVA with retention and comprehension scores as depended variables revealed no significant main effect for self-explanation prompts (*Wilk's* $\Lambda=0.999$, $F(2, 123)=0.08$, $p=.926$). Bayes factors indicated support for the null-hypothesis regarding retention ($BF_{10}=0.20$) and comprehension ($BF_{10}=0.20$). Figures 3 and 4 illustrate the descriptive data for the learning scores.

An additional MANOVA using the two types of comprehensions scores (thematically close vs. not close to the self-explanation prompts) revealed no significant main effect for self-explanation prompts (*Wilk's* $\Lambda=0.992$, $F(2, 123)=0.50$, $p=.610$). Bayes factors indicated support for the null-hypothesis (thematically close: $BF_{10}=0.20$, not close: $BF_{10}=0.27$). Hypothesis 2 is therefore not supported by the results.

Hypothesis 3 Interaction effect of navigation features and self-explanation prompts on learning performance.

A MANOVA with retention and comprehension scores as depended variables revealed no interaction effect (*Wilk's* $\Lambda=0.98$, $F(2, 123)=1.27$, $p=.283$). Bayes factors indicated support for the null-hypothesis regarding retention ($BF_{10}=0.04$) and comprehension ($BF_{10}=0.04$).

An additional MANOVA using the two types of comprehensions scores (thematically close vs. not close to the self-explanation prompts) revealed no interaction effect (*Wilk's* $\Lambda=0.98$, $F(2, 123)=1.33$, $p=.270$). Bayes factors indicated support for the null-hypothesis regarding questions that were thematically close to the self-explanation prompts ($BF_{10}=0.04$) and not thematically close ($BF_{10}=0.05$). Hypothesis 3 is therefore not supported by the results.

Hypothesis 4 Effect of quality of self-explanations on learning performance.

Responses to all self-explanation prompts were scored by two independent raters. Inter-rater reliability (two-way-mixed, absolute agreement) was sufficient ($ICC(3, k) = [0.77 - 0.98]$, $F(62, 62) = [8.21 - 99.61]$, $p < .001$). Self-explanation ratings did not differ based on the availability of navigation features ($t(52)=1.04$, $p=.301$, $d=0.28$). The Bayes factor could not differentiate between the null- and alternative hypothesis ($BF_{10}=0.43$). Regression analyses revealed that self-explanation ratings did not significantly predict retention

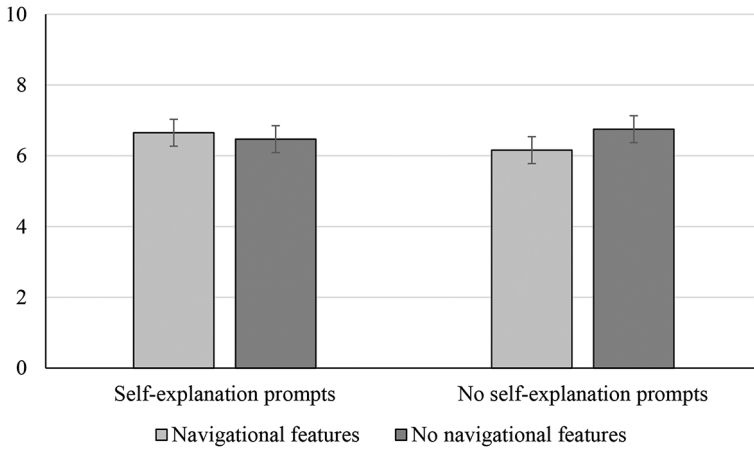


Fig. 3 Bar diagram for retention performance, M and SE are displayed

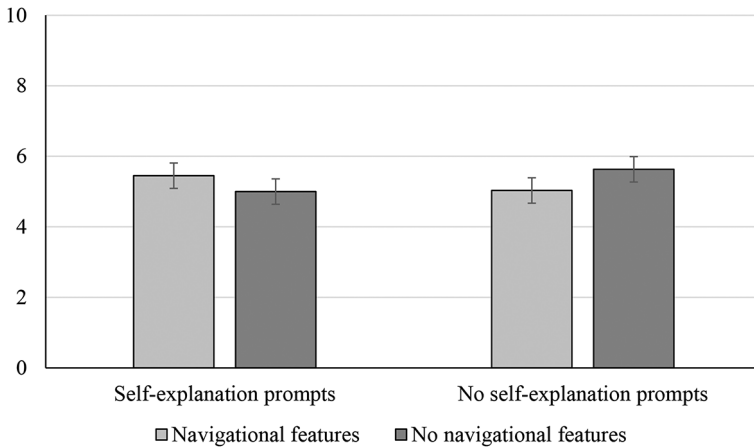


Fig. 4 Bar diagram for comprehension performance, M and SE are displayed

performance ($R^2=0.03$, $\beta=0.19$, $p=.180$). However, learners that produced better self-explanations showed improved comprehension performance ($R^2=0.09$, $\beta=0.29$, $p=.031$). However, this does not apply for comprehension questions that were thematically close to the self-explanation prompts ($R^2=0.05$, $\beta=0.21$, $p=.123$) and for questions that were not thematically close ($R^2=0.07$, $\beta=0.226$, $p=.062$). Hypothesis 4 is therefore partially supported by the results.

Hypothesis 5 Effect of navigation features and self-explanation prompts on cognitive load.

A MANOVA with intrinsic load, extraneous load, and germane load as depended variables revealed a significant main effect for self-explanation prompts ($Wilks' \Lambda=0.88$, $F(3, 122)=5.49$, $p=.001$) but not for navigation features ($Wilks' \Lambda=0.98$, $F(3, 122)=0.77$,

$p=.514$). Furthermore, no interaction effect was found ($Wilk's \Lambda=0.98$, $F(3, 122)=0.66$, $p=.577$).

Follow-up ANOVAS were conducted to get deeper insight into the significant main effect for self-explanation prompts. Results revealed a significant effect regarding intrinsic load ($F(1, 124)=8.40$, $p=.004$, $\eta_p^2=0.06$), a significant effect regarding extraneous load ($F(1, 124)=11.93$, $p<.001$, $\eta_p^2=0.09$) but no effect regarding germane load ($F(1, 124)=0.33$, $p=.564$, $\eta_p^2=0.003$). According to the descriptive data in Table 1, self-explanation prompts increased the perception of intrinsic and extraneous cognitive load. For germane load, Bayes factors indicated support for the null-hypothesis (effect of self-explanations: $BF_{10}=0.22$, effect of navigational features: $BF_{10}=0.28$, interaction: $BF_{10}=0.06$). Hypothesis 5 is therefore partially supported by the results. Descriptive plots illustrate the results for intrinsic cognitive load (Fig. 5), extraneous cognitive load (Fig. 6) and germane cognitive load (Fig. 7).

Hypothesis 6 Mediation effects.

Overall, mediation analyses revealed no significant indirect effects of self-explanation prompts and navigation features on learning performance. Intrinsic load did not act as mediator ($\beta = [-0.05 - -0.01]$, $SE = [0.02 - 0.05]$, $z = [-1.09 - -0.50]$, $p = [0.277 - 0.646]$) as well as extraneous load ($\beta = [-0.04 - 0.01]$, $SE = [0.02 - 0.06]$, $z = [-0.78 - 0.64]$, $p = [0.439 - 0.880]$) and germane load ($\beta = [-0.02 - 0.04]$, $SE = [0.03 - 0.04]$, $z = [-0.57 - 0.87]$, $p = [0.385 - 0.580]$).

Discussion

Previous studies have mostly reported positive effects of learning from interactive videos (see Ploetzner, 2024). It should be noted, however, that the studies reviewed by Ploetzner (2024) did not specifically investigate the effects of self-explanation prompts on retention and comprehension. Rather, they included different types of questions and different types

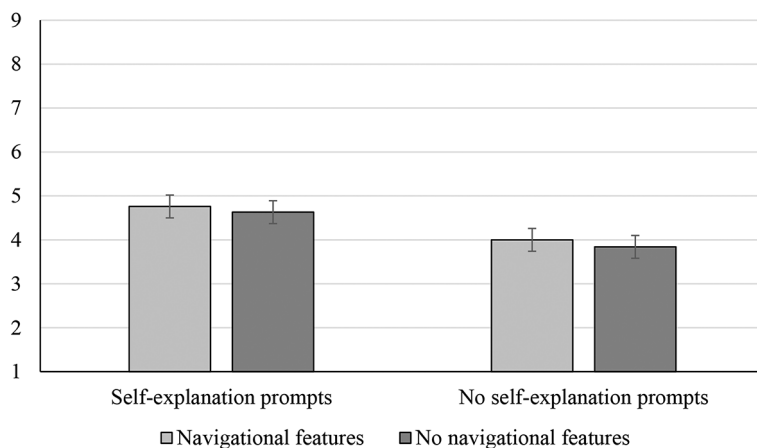


Fig. 5 Bar diagram for ICL, M and SE are displayed

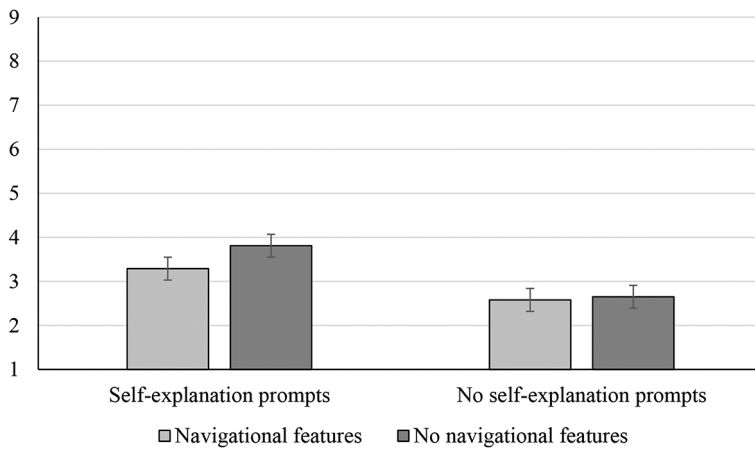


Fig. 6 Bar diagram for ECL, *M* and *SE* are displayed

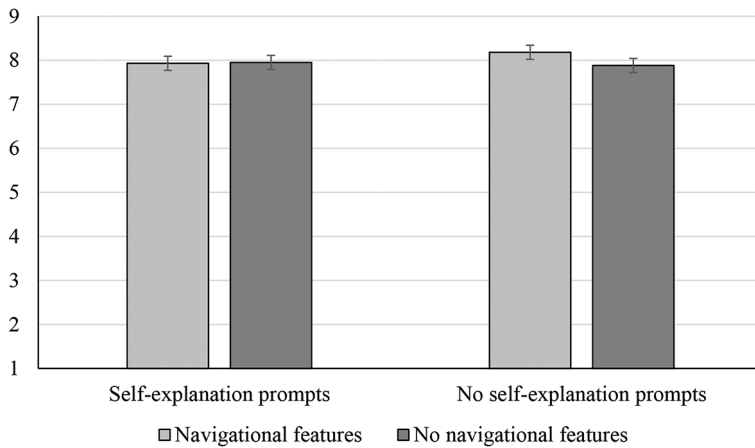


Fig. 7 Bar diagram for GCL, *M* and *SE* are displayed

of learning outcomes. Furthermore, many studies have confounded the effects of navigation features and enhanced features such as self-explanation prompts in instructional videos, and have not controlled for time on task. Similarly, studies of the effects of self-explanation prompts often do not control for time on task (for reviews, see Bisra et al., 2018; Dunlosky et al., 2013). Therefore, two aims of this study were (1) to disentangle the effects of navigation features and self-explanation prompts, and (2) to control for time on task.

In all groups, learning performance was moderate in terms of retention and poor in terms of comprehension. This was in spite of the fact that the subject matter was presented at a level of difficulty similar to that which is taught in secondary schools. Contrary to the first three hypotheses, neither navigation features nor self-explanation prompts improved retention or comprehension.

Navigation features did not benefit learning, even though learners were given instructions on how to improve their learning by using them strategically (see Lin et al., 2022). As observed in previous studies of learning from dynamic visualizations, it could be that learners nevertheless either made little use of navigation features (e.g., Hasler et al., 2007; Meyer et al., 2010; Rey & Diehl, 2010) or used them inefficiently (for a review, see Scheiter, 2022). For example, learners may lack metacognitive knowledge about how and when to use navigation features correctly, supporting the argument outlined by Renkl and Scheiter (2017). For example, learners may have paused the video at insignificant points or in the middle of a coherent explanation, thus failing to achieve the desired effect. To better understand the learners' use of navigation features, future research needs to capture this. One possibility is to record the learners' interaction behavior with screen recordings. However, analyzing the data obtained in this way would be very time-consuming. A more suitable option would be a video player that records the learners' interaction behavior as log files (e.g., Kuhlmann et al., 2024).

Self-explanation prompts did not improve learning either. This is the case even though the prompts were designed and used in a way that, based on previous research (see Bisra et al., 2018; Fiorella & Mayer, 2015), proved to be particularly advantageous: learners with little prior knowledge, acquisition of conceptual knowledge, focused prompts, and prompts that required drawing inferences. In addition, as with the navigation features, learners were informed about how self-explanation can support learning and encouraged to explain the learning content to themselves. Nevertheless, the quality of the self-explanations that were produced was only poor to moderate.

There are several reasons that may have contributed to the ineffectiveness of self-explanation in this study. First, the self-explanation prompts may have been too demanding for the learners. This could be due to a lack of prior knowledge as well as a lack of practice in generating self-explanations (cf. Dunlosky et al., 2013; Fiorella & Mayer, 2015). Second, as Fonseca and Chi (2011) pointed out, the need to type self-explanations may have a negative impact on their quality compared to verbal expression. In line with Olive (2012), writing itself may be effortful and thus result in lower response quality. In contrast, quality of answers might be improved if other answer formats (e.g., multiple choice) were chosen. However, this should be viewed with caution, as other processes may also be affected (free recall vs. recognition). Third, as discussed by Bisra et al. (2028), prompting for self-explanation might interfere with learners' self-regulated learning. That is, when prompted, learners might respond to the prompts but not to their self-diagnosed gaps in understanding. Fourth, some students' motivation may have been low. Some students may have participated in this study in order to receive a monetary reward. Thus, they were more extrinsically motivated to participate. As a result, they may not have been intrinsically motivated to engage deeply with the learning content, resulting in poor quality self-explanations. Fifth, Fiorella and Mayer (2015) highlighted that studies on the effectiveness of self-explanation have yielded particularly mixed results with respect to learning from dynamic visualizations. Finally, controlling for time on task may have further reduced the relative effectiveness of self-explanation (cf. Bisra et al., 2018; Dunlosky et al., 2013). In other words, re-watching the video several times can be as effective as trying to construct self-explanations.

However, controlling for learning time can lead to a dilemma. Balancing the learning time between conditions means that the learning time in the self-explanation conditions might be rather short, because of the text that had to be written. Thus, more learning time might

have led to better learning outcomes. On the other hand, the learning time for the conditions without self-explanations was already quite long. Learners had to watch the video several times which could lead to an illusion of understanding. More learning time would probably not have led to more learning, but only to more waiting. Therefore, future research needs to further investigate, under controlled conditions, which types of question and response formats are most effective in supporting learning from particularly dynamic visualizations such as interactive videos. Given that the relationship between learning time and learning performance is still not well understood (see Godwin et al., 2021), future research could also allow learners to self-regulate their learning time, assess learning time, and analyze how learning time affects learning, e.g., through moderator analyses.

The fourth hypothesis was partially supported. The quality of self-explanation was positively correlated with comprehension but not with retention. This may be due to the fact that the self-explanation prompts in this study were designed to encourage learners to make inferences and construct explanations rather than to remember the information presented. Nevertheless, average learning performance was no better than without self-explanation (cf. Hefter et al., 2023).

Regarding the fifth hypothesis, the availability of navigation features had no significant effect on cognitive load. This finding is in line with the suggestion above that learners may have made little use of navigation features. Self-explanation prompts had a significant effect on cognitive load. However, the pattern of effects only partially supports the hypothesis.

In all groups, learners experienced only moderate intrinsic load from the learning task. Given the poor to moderate learning performance, this may suggest that the learners underestimated the difficulty of the learning task, or even experienced an illusion of understanding, as has repeatedly been observed in previous studies of learning from dynamic visualizations (e.g., Kühl et al., 2011; Paik & Schraw, 2013). Such misapprehensions may also underlie the finding reported by Fiorella and Mayer (2015) that self-explanation studies have yielded mixed results regarding learning from dynamic visualizations. Intrinsic load was significantly increased by self-explanation prompts, possibly indicating that they made learners more aware of the learning tasks' complexity.

In contrast, learners rated the germane load induced by the learning task as rather high. This may reflect the fact that learners perceived themselves as making a lot of effort to learn, even though the learning task was perceived as achievable. Contrary to our hypothesis, self-explanation prompts did not increase germane load any further.

Extraneous load was rather low in all groups. As predicted, it was significantly increased by self-explanation prompts. There could be two reasons for this. First, self-explanation prompts pause the video, interrupting the flow of information. After providing answers, learners may have to re-establish their processing of the video, which may interfere with knowledge construction. The processes involved in dealing with multiple interruptions in a video can be seen as a form of extraneous load that can lead to an incomplete mental model of the material presented. Second, and more specifically for the material used in this study, the presentation of the self-explanation prompts changed the screen design (see Fig. 2). While a question-and-answer area was presented on the left side of the screen, the video was presented in a smaller size on the right side of the screen. After a response was given, the video was shown again in full-screen mode until the next self-explanation prompt was presented (see Fig. 1). These changes in screen design required the learners to repeatedly re-

orient themselves in the user interface, which probably contributed to the perceived extraneous load. A consistent user interface could help to keep extraneous load low.

Finally, the exploration of whether cognitive load mediated the effects of navigation features and self-explanation prompts yielded no significant results. As neither navigation features nor self-explanations had an effect, this is not surprising.

Conclusions

While recent research has reported positive effects of learning from interactive videos enriched with questions (for an overview see Ploetzner, 2024), the current study does not support this. Neither navigation features nor self-explanation prompts enhanced learning, but neither did they hinder it. Thus, enriching instructional videos according to the ICAP framework (Chi & Wylie, 2014) so that learning shifts from receptive to active and constructive learning is not generally beneficial. Rather, the quality of the self-explanations and the control of the learning time seem to play an important role in the interpretation of this result.

It is important to emphasize how interactive videos are actually used by learners. If the quality of the self-explanations is rather low due to the difficulty of the content, there will be no learning effects beyond the effects of restudying. Furthermore, if navigation features are not used or are used ineffectively, there will be no positive effects on learning, even if previous research suggests otherwise (Rey et al., 2019). Thus, the positive effects of interactive video on learning as summarized by Ploetzner (2024) may not be generalizable and moderator variables should be considered.

Future research should therefore focus on boundary conditions. Systematic research could outline how interactive elements are processed using eye-tracking and log-file analysis. Finally, the complexity and focus of the self-explanation prompts, the position of the prompts in the video, and the time allotted to formulating self-explanations may be important influencing factors.

Author contribution Maik Beege: Conceptualization; Methodology; Investigation; Data Curation; Formal Analysis; Validation; Project Administration; Writing - Original Draft; Writing - Review & Editing; Supervision. Rolf Ploetzner: Conceptualization; Methodology; Investigation; Data Curation; Formal Analysis; Validation; Project Administration; Writing - Original Draft; Writing - Review & Editing; Supervision.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval All procedures were performed in full accordance with the ethical guidelines of the German Psychological Society and the APA Ethics code with written informed consent from all subjects. Since the experiments constitutes non-medical low risk research, no special permission by an ethics committee is required for psychological research in the Institute for Psychology of the Freiburg University of Education as well as in Germany. At the beginning of the experiment, participants were informed that the data of this study will be used for research purposes only and that all data are collected anonymously. Thus, no identify-

ing information was collected. Participants who prematurely stopped the survey were not included in the analyses and all of their data were deleted from the dataset.

Conflict of interest I declare that I and all Co-Authors have no conflict of interest.

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